FetalAI: Using Machine Learning to Predict and Monitor Fetal Health

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1. Introduction

Fetal health monitoring is a critical aspect of prenatal care. Early detection of fetal distress or abnormalities can significantly improve maternal and neonatal outcomes. This project, FetalAI, aims to leverage machine learning techniques to predict fetal health conditions based on medical parameters. The system provides an intelligent solution that assists healthcare professionals in decision-making and enables early intervention.

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2. Problem Statement

Traditional fetal monitoring methods rely heavily on manual interpretation, which can lead to delayed diagnosis and human error. There is a need for an automated system that can analyze medical data efficiently and provide accurate predictions of fetal health conditions such as normal, suspect, or pathological.

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3. Objectives

The main objectives of this project are:

- To build a machine learning model for fetal health prediction
- To analyze medical data and identify risk factors
- To develop a web-based application for real-time prediction
- To assist healthcare providers in early detection and intervention

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4. Scope of the Project

The project focuses on predicting fetal health using structured medical data. It includes data preprocessing, model training, evaluation, and deployment through a web interface. The system is designed for educational and prototype purposes and can be extended for real-world healthcare applications.

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5. Literature Survey

Various studies have explored the use of machine learning in healthcare, especially in fetal monitoring using Cardiotocography (CTG) data. Algorithms such as Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines have been widely used. Among these, ensemble methods like Random Forest have shown high accuracy and robustness in classification tasks.

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6. Dataset Description

The dataset used in this project is the Fetal Health Classification dataset. It contains multiple features related to fetal heart rate, uterine contractions, and variability measures. The target variable classifies fetal health into three categories:

- Normal
- Suspect
- Pathological

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7. Data Preprocessing

Data preprocessing involves handling missing values, selecting relevant features, and scaling the data. Feature selection was performed to ensure that only important parameters are used for model training, improving both performance and efficiency.

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8. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the dataset. Statistical summaries, distribution plots, and correlation heatmaps were used to identify relationships between variables and detect patterns in the data.

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9. Feature Selection

Relevant features were selected based on correlation analysis and domain understanding. This step ensures that the model is trained using only the most significant features, reducing complexity and improving prediction accuracy.

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10. Model Building

Multiple machine learning models were implemented, including:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors

These models were trained and tested using the prepared dataset.

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11. Model Evaluation

The performance of each model was evaluated using metrics such as:

- Accuracy
- Precision
- Recall
- F1-score

Among all models, Random Forest achieved the highest accuracy and was selected as the final model.

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12. Handling Imbalanced Data

To address class imbalance in the dataset, SMOTE (Synthetic Minority Over-sampling Technique) was applied. This technique improves the model's ability to correctly classify minority classes.

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13. Model Deployment

The trained model was saved using the pickle library and deployed using the Flask web framework. A user-friendly interface was developed to allow users to input medical parameters and receive predictions in real time.

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14. System Architecture

The system consists of three main components:

- Frontend (HTML/CSS) for user interaction
- Backend (Flask) for processing requests
- Machine Learning Model for prediction

The user inputs data through the UI, which is processed by the model, and the result is displayed back to the user.

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15. Results and Discussion

The Random Forest model achieved high accuracy and performed well in classifying fetal health conditions. The system successfully provides predictions based on user input, demonstrating the effectiveness of machine learning in healthcare applications.

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16. Conclusion and Future Work

The FetalAI system demonstrates how machine learning can be used to predict fetal health and assist in early diagnosis. Future improvements may include:

- Integration with real-time IoT devices
- Use of deep learning models
- Deployment on cloud platforms
- Enhancement of UI/UX for better usability

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17. Outputs

FetalAI - Predict Fetal Health
120
0.002
0.01
0.0001
240
0.0
0.0
200
Predict
Result: Normal

FetalAI - Predict Fetal Health

Baseline Value

Accelerations

Fetal Movement

Uterine Contractions

Light Decelerations

Severe Decelerations

Prolongued Decelerations

Abnormal Short Term Variability

Predict

Result: Pathological